

Teaching Exam

Subject – Computer Science

Topic – DBMS



Lecture No. – 9

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Topics to be Covered

Topic



- 1:- Normalization ✓
- 2:- 1- NF (Normal Form) ✓
- 3:- Closer method to find candidate key ✓
- 4:- Some example related to closer method ✓
- 5:- Functional Dependency ✓
- 6:- Types of Functional Dependency ✓
- 7:- Trivial Function Dependency ✓
- 8:- Non-Trivial Functional Dependency ✓
- 9:- Semi Non-Trivial Function Dependency ✓
- 10- Multivalued Functional Dependency ✓
- 11- Transitive Functional Dependency ✓
- 12- Full and Partial Functional Dependency ✓
- 13- Conclusion. ✓



Topic :Normalization



TEACHING
WALLAH

Normalization is the process of organizing the data and the attributes of a database. It is performed to reduce the data redundancy in a database and to ensure that data is stored logically. Data redundancy in DBMS means having the same data but at multiple places. It is necessary to remove data redundancy because it causes anomalies in a database which makes it very hard for a database administrator to maintain it.

Row Wise Redundancy:-

SID	SNAME	AGE
1	RAVI	20
2	SURYA	23
1	RAVI	21



Topic : Normalization



COLLEGE MANAGEMENT

STUDENT

COURSE

FACULTY

SID	SNAME	CID	CNAME	FID	FNAME	SALARY
1	RAM	C1	DBMS	F1	JOHN	30000
2	SURYA	C2	JAVA	F2	RAVI	40000
3	NITIN	C1	DBMS	F1	JOHN	30000
4	AMRIT	C1	DBMS	F1	JOHN	30000
		C10	CC			

Column wise
Redundancy:-



Topic :Normalization

Why Do We Need Normalization?

As we have discussed above, normalization is used to reduce data redundancy. It provides a method to remove the following anomalies from the database and bring it to a more consistent state:

A database anomaly is a flaw in the database that occurs because of poor planning and redundancy.

1:-**Insertion anomalies**: This occurs when we are not able to insert data into a database because some attributes may be missing at the time of insertion.

2:-**Updation anomalies**: This occurs when the same data items are repeated with the same values and are not linked to each other.

3:-**Deletion anomalies**: This occurs when deleting one part of the data deletes the other necessary information from the database.



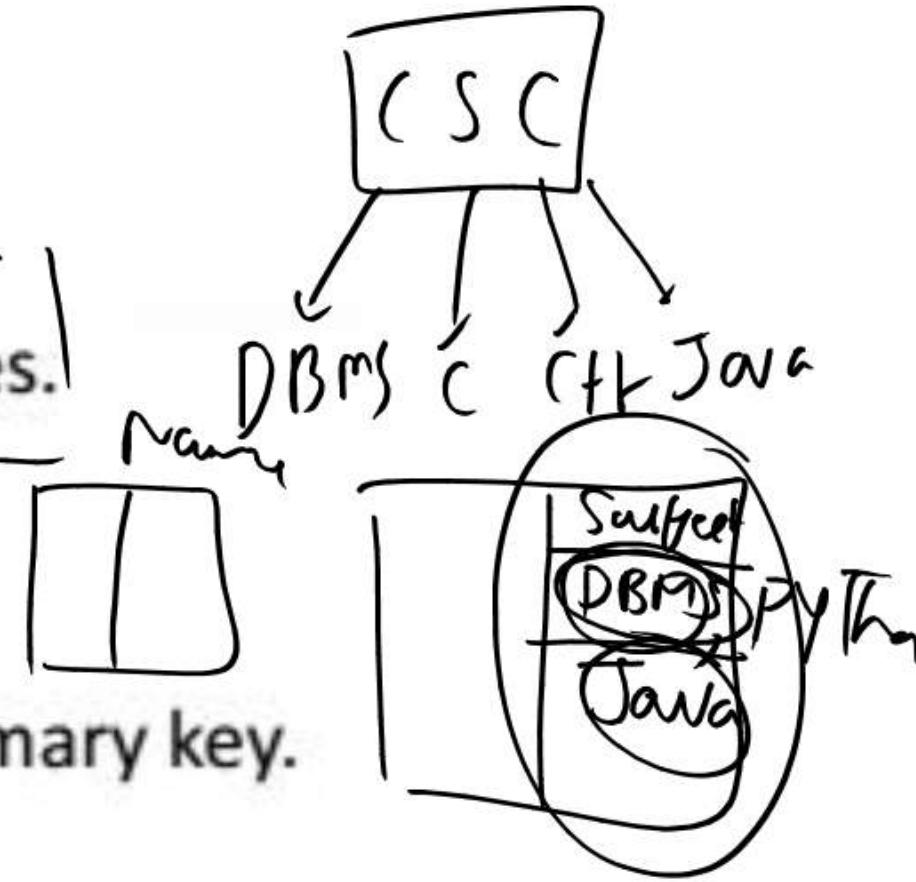
Topic :1-NF

If a relation contains a composite or multi-valued attribute, it violates the first normal form, or the relation is in the first normal form if it **does not contain** any composite or multi-valued attribute. A relation is in first normal form if every attribute in that relation is single-valued attribute.

A relation (table) is said to be in First Normal Form (1NF) if:

- All the attributes (columns) contain only atomic (indivisible) values.
- Each column contains values of a single type.
- Each record (row) is unique, meaning it can be identified by a primary key.
- There are no repeating groups or arrays in any row.

Composite Attribute $\nexists$ 1NF





Topic :1-NF

It is Not in 1 NF

ROLL NO	NAME	COURSE
1	RAVI	C/C++
2	HARI	JAVA
3	OHM	C/DBMS

METHOD- 1ST

COMPOSITE KEY (ROLL NO+COURSE), HERE THERE IS NO PRIMARY KEY

ROLLNO	COURSE	NAME
1	C	RAVI
1	C++	RAVI
2	JAVA	HARI
3	C	OHM
3	DBMS	OHM

+ Multivalued



Topic :1-NF

METHOD-2ND

Here Primary key is ROLLNO but here chance of null value comes more

X Multivalued
- Composite Attribute

ROLLNO	NAME	COURSE1	COURSE2
1	RAVI	C	C++
2	HARI	JAVA	NULL
3	OHM	C	DBMS



Topic :1-NF

METHOD- 3RD (BEST WAY TO SOLVE 1 NF)

PK

<u>ROLLNO</u>	NAME
1	RAVI
2	HARI
3	OHM

FK

<u>ROLL NO</u>	COURSE
1	C
1	C++
2	JAVA
3	C
3	DBMS

BASE TABLE ----> PRIMARY KEY -ROLL NO

COMPOSITE KEY-(ROLLNO+COURSE)

FOREIGN KEY- ROLL NO

(Child) fk is subset
of PK (Base)

Multivalued
Composite
Attribute

Composite
Key



Topic :Closer Method

Closure of an Attribute Set-

- The set of all those attributes which can be functionally determined from an attribute set is called as a closure of that attribute set.
- Closure of attribute set $\{X\}$ is denoted as $\{X\}$

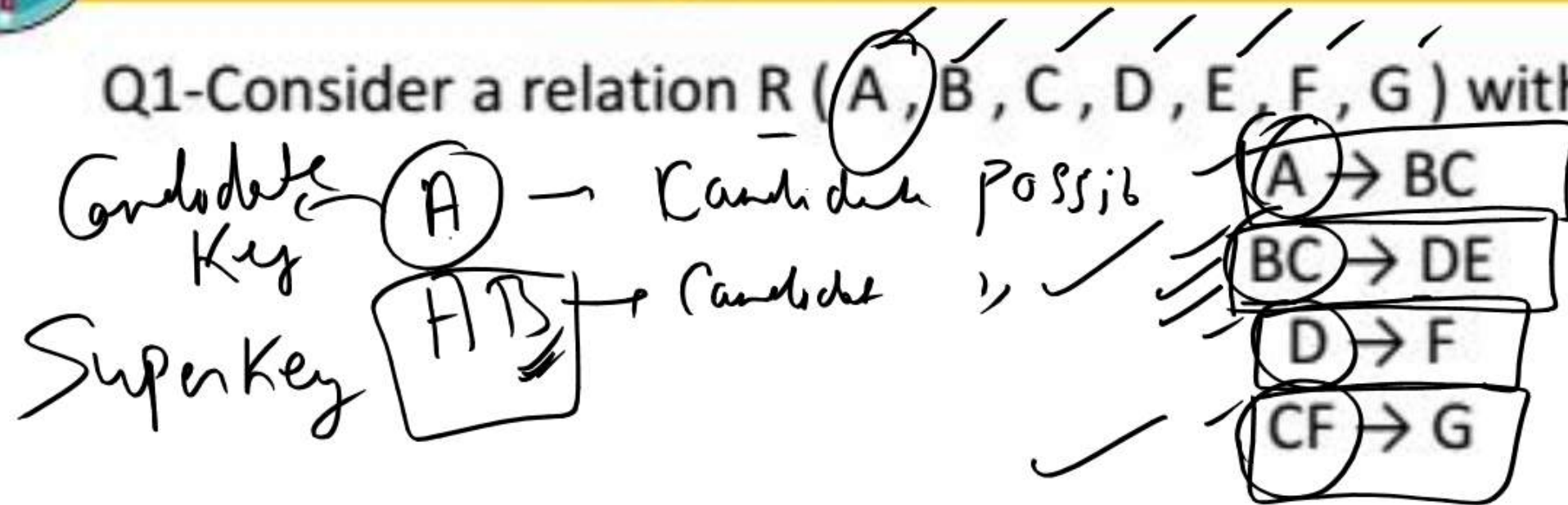
Steps to Find Closure of an Attribute Set-

- 1:-Add the attributes contained in the attribute set for which closure is being calculated to the result set
- 2:-Recursively add the attributes to the result set which can be functionally determined from the attributes already contained in the result set.



Topic : Closer Method

Q1-Consider a relation R (A, B, C, D, E, F, G) with the functional dependencies-



$$A \xrightarrow{\text{closer}} A^+$$

$$A^+ = \{A, B, C, D, E, F, G\}$$

Now, let us find the closure of some attributes and attribute sets-

$$\begin{aligned} A^+ &= \{A\} \\ &= \{A, B, C\} && \text{(Using } A \rightarrow BC) \\ &= \{A, B, C, D, E\} && \text{(Using } BC \rightarrow DE) \\ &= \{A, B, C, D, E, F\} && \text{(Using } D \rightarrow F) \\ &= \{A, B, C, D, E, F, G\} && \text{(Using } CF \rightarrow G) \end{aligned}$$

Thus,

$$A^+ = \{A, B, C, D, E, F, G\}$$

Handwritten notes for other attributes:

- $B^+ = B$ (crossed out)
- $C^+ = C$ (crossed out)
- $D^+ = D, F$ (crossed out)

Note:- If any Attribute like A here closer to all attribute including itself that A attribute is possible to be Super key or Candidate Key.



Topic :Closer Method

Super Key-

- If the closure result of an attribute set contains all the attributes of the relation, then that attribute set is called as a super key of that relation.
- Thus, we can say-
“The closure of a super key is the entire relation schema.”

Example-

In the above example,

- The closure of attribute A is the entire relation schema.
- Thus, attribute A is a super key for that relation.**



Topic :Closer Method

Candidate Key-

- If there exists no subset of an attribute set whose closure contains all the attributes of the relation, then that attribute set is called as a candidate key of that relation.

Example-

In the above example,

- No other attribute contains all the attributes of the relation, except A attribute.
- Thus, attribute A is also a candidate key for that relation.



Topic : Closer Method

Question-2 Consider the given functional dependencies-

$$AB \rightarrow CD \quad \checkmark$$

$$AF \rightarrow D \quad \checkmark$$

$$DE \rightarrow F \quad \checkmark$$

$$C \rightarrow G \quad \checkmark$$

$$F \rightarrow E \quad \checkmark$$

$$G \rightarrow A \quad \checkmark$$

$$\underline{R(A, B, C, D, E, F, G)}$$

Which of the following options is false?

(A) $\{CF\}^+ = \{A, C, D, E, F, G\}$ ✓

(B) $\{BG\}^+ = \{A, B, C, D, G\}$ ✓

(C) $\{AF\}^+ = \{A, C, D, E, F, G\}$ ✗

(D) $\{AB\}^+ = \{A, C, D, F, G\}$ ✗

$$C^+ = \{C, F, G, E, A, D\}$$

$$BG^+ = \{B, G, A, C, D\} \quad \text{✗}$$

$$AF^+ = \{A, F, D, E\}$$

$$AB^+ = \{A, B, C, D, G\} \quad \text{✗}$$

THANK YOU